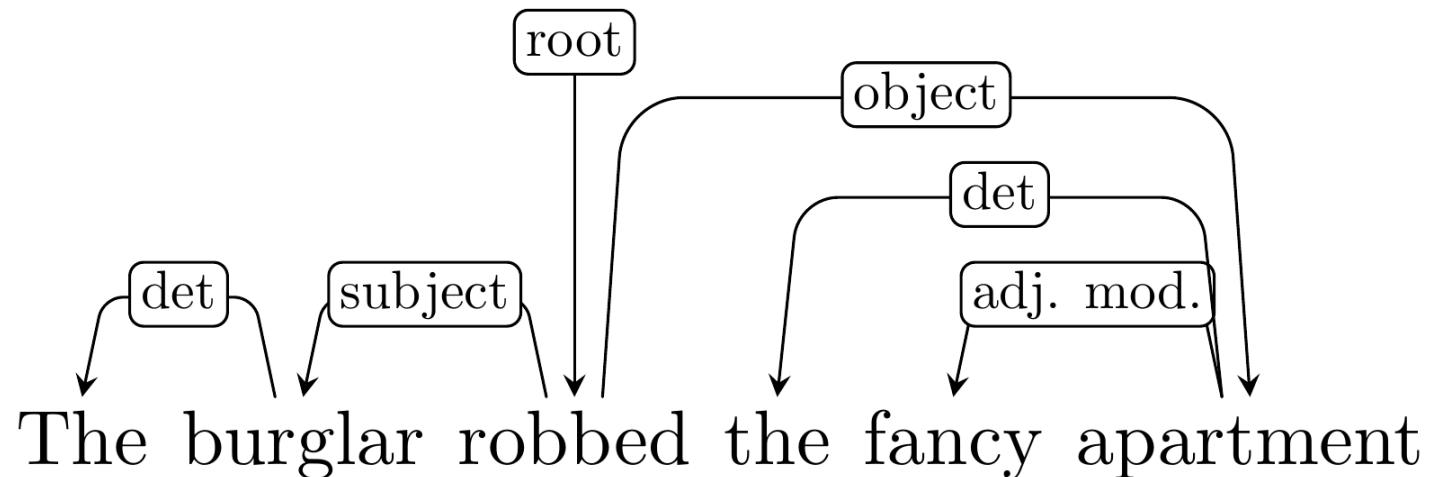


Basic Units: Syntactic Heads

- The notion of a *head* is used in a wider context than noun phrases
- The verb is considered the head of a clause:
 - Arguments and modifiers are considered its *dependents*
 - **John** ran **home** (“ran” is the head of the clause, “John” and “home” are its dependents)
 - **Bill stupidly** answered **the officer’s question** (“answered” is the head, “Bill”, “stupidly” and “the officer’s question” are its dependents)
- Some schemes view all syntactic relations as head-dependent relations (*dependency schemes*)

Dependency Parses

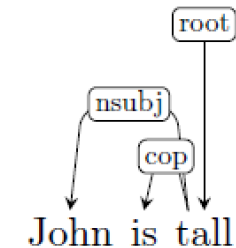
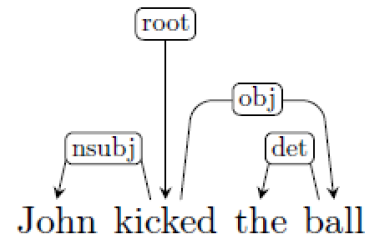
- Represent Syntax as head-dependent relations between words
 - Nodes are words, edges are relations
 - The arguments and modifiers of a word are its children
 - Edge labels mark the type of relation
 - The verb of the main clause is the root



Example Annotations in *Universal Dependencies* Format

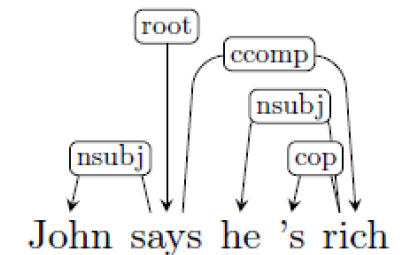
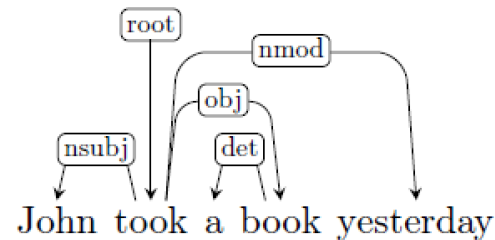
- Transitive clauses:

- John kicked the ball
- John took a book yesterday



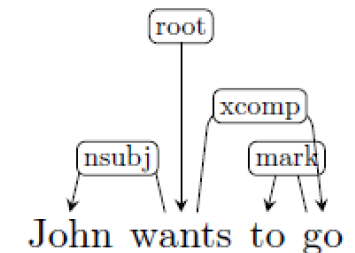
- Copula clauses:

- John is tall



- Complement clauses:

- John says he's rich
- John wants to go



<http://universaldependencies.org/>

<http://universaldependencies.org/u/dep/index.html>

Example Annotations in *Universal Dependencies* Format

- Relative Clauses:
 - The dog that ate my homework
- Subordinate clauses:
 - After John came home, he took a shower
- Coordination:
 - John and Mary got married

